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Reflective Journal: IIoT Protocols Project

1. Introduction

This project focused on designing and simulating an Industrial Internet of Things (IIoT) sensor network using three communication protocols: MQTT, CoAP, and OPC UA. The objective was to simulate real-time temperature and humidity data transmission and visualize it dynamically. My personal goal for this project was to understand how industrial communication protocols operate in practical environments and how real-time data flows from sensors to visualization layers. I aimed to strengthen my understanding of broker-based architectures and real-time monitoring systems, which are essential in industrial automation and smart manufacturing environments.

2. Personal Contributions

In this project, I primarily supported the data pipeline and visualization layer. My responsibilities included assisting with data extraction from the MQTT broker, supporting the implementation of real-time charting using Matplotlib, refining rolling data handling to ensure smooth updates, testing MQTT subscription flows, validating JSON payload formatting, and capturing screenshots and video demonstrations for the visualizations folder. I collaborated with the team to ensure the visualization accurately reflected sensor updates and maintained consistency across protocols for comparison.

3. Learning Outcomes

Through this lab, I developed a deeper understanding of three industrial communication protocols. MQTT operates on a lightweight publish-subscribe model using a broker, making it ideal for IoT telemetry systems. CoAP uses a request-response model over UDP optimized for constrained devices and low-power environments. OPC UA provides a structured and secure industrial communication framework with hierarchical address spaces and writable nodes used in automation systems. A key takeaway is that different protocols serve different industrial needs, and selecting the correct protocol depends on system requirements such as scalability, security, and computational constraints.

4. Challenges and Solutions

One challenge involved correctly parsing incoming JSON data for visualization without runtime errors. This was addressed by validating payload formatting and improving the parsing logic. Another challenge was maintaining smooth real-time graph updates without performance degradation.

Implementing a rolling data window limited the dataset to recent entries and improved performance. Working across multiple protocols also required ensuring consistent data structures, which improved my debugging and asynchronous system troubleshooting skills.

5. Future Applications

The knowledge gained from this project can be applied to industrial monitoring systems, predictive maintenance platforms, IoT dashboards, and smart factory environments. Future improvements could include implementing TLS security for MQTT, deploying a cloud-based broker, integrating a web dashboard using frameworks like Flask or Streamlit, and conducting a formal latency comparison between protocols. This project reinforced my interest in AI-integrated industrial systems and IIoT architectures within the AI and Robotics field.